Title: Spread Locator: A Statistical Distribution Analysis Model
Duration: 6 Hours
Type: Theory + Practical

🎯 Objective:

To help students understand and apply the concepts of **probability distributions and spread analysis** on a given dataset. This will test their theoretical understanding and practical application of distribution types, Q-Q plots, statistical transformations, and probability functions.

Project Scenario:

You are a data analyst working for an **e-commerce platform** that wants to understand customer purchase behaviors by analyzing **daily transaction amounts** of customers. Management is particularly interested in knowing whether the transactions follow certain statistical distributions, how to handle skewed data, and what probability insights can be derived.

Your task is to apply **statistical distribution concepts and transformations** to derive insights on transaction behaviors.

🧩 Dataset Structure (Download Dataset)

Field Name	Data Type	Description
transaction_id	UUID/String	Unique identifier for each transaction
customer_id	UUID/String	Unique identifier for each customer
transaction_amount	Float	Total amount of the transaction (₹)

transaction_date	Date	Date of the transaction
transaction_count	Int	Number of transactions made by a customer in a given week
region	String	Customer's geographic region (North, South, East, West)
transaction_status	String	Whether transaction was successful or failed (Success/Fail)

📌 Task Checklist

🔹 **Part A - Theoretical Foundation (Short Notes & Explanation):**

Write a detailed note covering:

1. What is Statistical Distributions?
2. What is a Q-Q Plot and why is it used?
3. Difference between Discrete and Continuous Distributions.
4. What is Bernoulli Distribution?
5. What is Binomial Distribution?
6. Explain Log-Normal Distribution.
7. Explain Power Law Distribution.
8. What is Box-Cox Transform?
9. Explain Poisson Distribution with an example.
10. What is Z-score Probability?
11. Differentiate Probability Density Function (PDF) and Cumulative Distribution Function (CDF).

🔹 **Part B - Data Analysis & Testing Tasks:**

Perform the following using Python (NumPy, Pandas, SciPy, Statsmodels, Matplotlib, Seaborn):

1. Fit the data to Bernoulli and Binomial distributions (transaction occurrence & weekly count).
2. Fit the data to Poisson distribution (number of transactions per day).
3. Model transaction amounts using Log-Normal and Power Law distributions.
4. Generate and interpret a **Q-Q Plot** to test normality.
5. Apply **Box-Cox Transform** to stabilize variance.

6. Calculate **Z-scores** for transaction amounts and compute probability of transactions exceeding ₹5000.
7. Plot and interpret **PDF and CDF** for transaction amounts.
8. Conclude which distribution best fits the dataset and justify insights for decision-making.

✅ Submission Guidelines

- Include practical implementation in Jupyter Notebook, Excel/Sheets, or screenshots.
- Label all charts clearly and write short interpretations under each result.
- GitHub Repository:
  - Create a GitHub repository to host your project.
  - Upload your project files, including source code and documentation.
  - Add a document (PDF) explaining theory concepts with definitions.
  - Ensure that you provide a clear and descriptive README.md file.